

ELLIOT HAWKINS

elliott.hawkins@icloud.com • [LinkedIn](#) • [Portfolio](#)

EDUCATION

Master of Science in Electrical Engineering and Computer Science; Electrical Engineering Emphasis Fowler School of Engineering, Chapman University, Orange, CA	May 2027
Bachelor of Science in Electrical Engineering; Minor: Music Technology Fowler School of Engineering, Chapman University, Orange, CA	May 2026 3.92 GPA

RELEVANT COURSES

Electronics and Circuits II | Control Systems | Electromagnetics I | Digital Logic Design II | Systems Programming
Engineering Mathematics | Microelectronics | Robotics | Digital Signals and Filters | Linear Algebra and Differential Equations

TECHNICAL SKILLS

Circuit Analysis | PCB Design and Fabrication | Python and C/C++ | Embedded Systems | Soldering | Laser Cutting | 3D Printing
Signal Processing | Arduino | 3D CAD | Project Management | MatLab | System Verilog | Waveform Analysis | Adobe Suite | CNC

EXPERIENCE

Student Makerspace Lead Employee, (DCILab) Chapman University	September 2022 - (Lead)August 2025 - Present
- Guided and supported student projects by providing expertise in: - PCB CNC and milling machines Circuit analysis and testing 3D printing and resin printing Laser cutting Soldering Communication and organization Poster and sticker printing Wood and metal fabrication Post-production Project Management - Developed and maintained comprehensive, streamlined instructions for equipment operation and sample project implementation - Developed and directed hands-on interactive workshops in: - Laser Cutting (File set-up and process) Soldering (Introduction to PCB soldering) Introduction to 3D printing (printing process) Introduction to Resin Printing (printing process) - Project workshops: “Useless Machine” (Soldering electronics laser cutting skills) Anti-Gravity figure (3D modeling printing)	
Teacher’s Assistant, Electromagnetics II (EENG-430) Chapman University	Fall 2024
- Modeled antennas using Ansys HFSS software for advanced electromagnetic field simulations - Developed curriculum support materials to assist the professor in enhancing student learning - Conducted office hours to provide personalized software assistance to students	
Teacher’s Assistant, Introduction to PCB design and Fabrication (CPSC-298) Chapman University	Fall 2025
Guided students through PCB design and fabrication on LPKF tools: ProtoMat and ProtoLaser	

PROJECTS

Target Retrieval System for Gun Ranges (Independent Contracting)	December 2025-Present
- Designed industrial motor control system with touchscreen HMI, PLC integration, and three-phase motor command output - Programmed industrial PLCs for real-time motor control logic including speed and operational sequencing - Designed custom PCB for signal conditioning and voltage regulation to interface with motor controller with high efficiency - Design a custom User Interface on the HMI utilizing TTL UART RX/TX to Modbus 485 serial communication conversion - Configured multi-voltage power distribution from 110V AC mains to regulated DC rails, integrating circuit breakers and existing power supply hardware for safe system operation	
Remote Controlled Life-Sized Functional R2-D2 Replica from Star Wars (Robotics Club)	February-August 2025
- Project lead and head of operations in all sub-groups - Designed and fabricated custom PCBs for R2-D2’s lighting systems with custom sequences on programmable integrated circuits - Collaborated with teammates to fabricate a control system that integrated all R2-D2 functionality, including movement and sound - Utilized Raspberry Pi’s networking capabilities to fabricate a custom controller equipped with a user interface - Gained problem-solving experience in every aspect of the project, including post-processing, electrical, and software areas	
“Useless Machine” Workshop (DCI workshop)	January - May 2024
- Developed a machine to contact and engage a switch with a motor-controlled plank when initiated by the user - Utilized electronic components that operated without requiring programming, microcontrollers, or Arduino-based systems - Expanded knowledge of electronic housing, fabrication, engineering process with project management, and circuit development - Streamlined the manufacturing process and facilitated workshops for students to assemble their own	

ACHIEVEMENTS & ACTIVITIES

Chapman Robotics Club, Secretary	Fall 2024 - Present
Society of Manufacturing Engineers, Secretary	Spring 2025 - Present
Fowler School of Engineering Deans Scholarship, Recipient	Fall 2022 - Present
Chapman University Provost List, Member	Spring 2023, Spring 2024, Fall 2024
Chapman University Computer Science Club, Member	Fall 2022 - Present
Chapman University Panther Game Development, Member	Fall 2024 - Present
World Cube Association, Staff Member, and Competitor	2017 - Present